Expectation values and operators

$$\langle x \rangle = \int dx \; P(x,t) \cdot x = \int dx \; \psi^*(x,t) \cdot x \, \psi(x,t)$$

in general:

$$\langle g(x) \rangle = \int dx \; \psi^*(x,t) \; g(x) \; \psi(x,t)$$

parameter

Def: linear operator $\hat{Q}$

linear mapping $\quad \hat{Q}: \; f(x) \rightarrow f'(x)$

$$\hat{Q} f = f'$$

$$(\hat{Q} f)(x) = f'(x)$$

linear:

$$\hat{Q}(\alpha \cdot f + \beta \cdot g) = \alpha \hat{Q} f + \beta \hat{Q} g = \alpha f' + \beta g'$$

position operator

$\hat{x} \equiv$ multiplication by x

$$\hat{x}: \; \psi(x,t) \rightarrow x \cdot \psi(x,t)$$

$$\langle x \rangle = \int dx \; \psi^*(x,t) \; \hat{x} \, \psi(x,t)$$

Def:

$$\boxed{\langle Q \rangle \equiv \int dx \; \psi^*(x,t) \, (\hat{Q} \psi)(x,t)}$$

Physical quantities $\Leftrightarrow$ operators !!

momentum operator

$$\hat{p} = ??$$

$$e^{-i\omega\cdot t}\, e^{ik\cdot x} \qquad p = \hbar\cdot k$$

$$\underset{\hbar\cdot k}{\underset{\vee}{p}}\; e^{-i\omega\cdot t}\, e^{ik\cdot x} = \underbrace{\hbar\,\frac{1}{i}\frac{\partial}{\partial x}}_{\hat{p}}\, e^{-i\omega\cdot t}\, e^{ik\cdot x}$$

$$\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$$

momentum operator !

$$\hat{p}\,\psi(x,t) = \frac{\hbar}{i}\frac{\partial\psi}{\partial x}$$

$$\langle p \rangle = \int dx\, \psi^*(x,t)\, \frac{\hbar}{i}\frac{\partial\psi}{\partial x}(x,t)$$

energy ?

$$\hbar\omega\psi \rightarrow i\hbar\,\partial_t\psi$$

$$\langle E \rangle = \int dx\, \psi^*(x,t)\, i\hbar\,\partial_t\psi(x,t)$$

Hamilton operator

$$\langle E \rangle = \int dx\, \psi^* \, i\hbar\, \partial_t \psi$$

$$\overset{!}{=} \int dx\, \psi^* \underbrace{\left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\,\psi \right)}_{= \underbrace{\left(\frac{\hat{p}^2}{2m} + \hat{V} \right)}_{\hat{H}} \cdot \psi}$$

$\hat{H}$ = Hamilton operator

$\sim$ energy !

$$\langle E \rangle = \langle H \rangle = \left\langle \frac{\hat{p}^2}{2m} + \hat{V} \right\rangle$$

kinetic energy

potential energy

Schrödinger equation :

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi}$$ time dependent S.E.

$$\Rightarrow \quad \boxed{E\varphi = \hat{H}\varphi}$$ time independent S.E.

Physical quantities $\leftrightarrow$ Hermitian operators $\hat{O}$

$\sim$ Hermitian matrices

- eigenvalues and eigenfunctions

$$\hat{O} \cdot \varphi_n(x) = \lambda_n \varphi_n(x)$$

eigenfunction

eigenvalues (real !)

Exchange relation

Take any wave function $\psi(x)$ $(t=0)$

$$\hat{x}\,\psi(x) = x\,\psi(x)$$

$$\Rightarrow \quad \hat{p}\,\hat{x}\,\psi = \frac{\hbar}{i}\frac{\partial}{\partial x}\big(x\,\psi(x)\big) = \frac{\hbar}{i}\psi(x) + \underbrace{x\,\frac{\hbar}{i}\frac{\partial\psi}{\partial x}}_{\hat{x}\,\hat{p}\,\psi}$$

$$(\hat{p}\,\hat{x} - \hat{x}\,\hat{p})\,\psi = \frac{\hbar}{i}\psi \qquad \text{for any } \psi$$

$$\Rightarrow \quad \boxed{[\hat{p},\hat{x}] = \hat{p}\,\hat{x} - \hat{x}\cdot\hat{p} = \frac{\hbar}{i}}$$

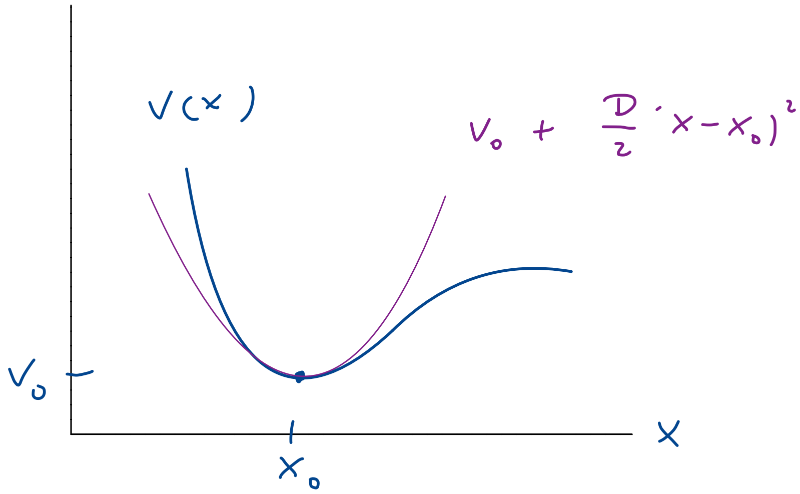
commutator

Heisenberg's exchange relation

This turns out to be the heart of Q.M. !

$\forall$ follows from this, algebraically !!!

Harmonic oscillator



close to potential minimum

$$V(x) = V_0 + \frac{D}{2}(x-x_0)^2 + \dots$$

we set $V_0, x_0 \to \emptyset$, for simplicity

Classical problem:

$$V = \frac{D}{2}x^2 \Rightarrow \text{force} \quad F = -\frac{\partial V}{\partial x} = -D \cdot x$$

Newton:

$$m \cdot \ddot{x} = -D \cdot x$$

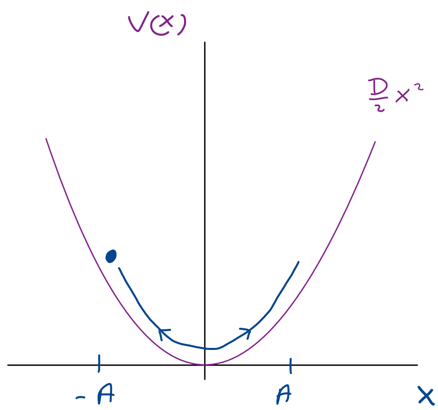
$$\Rightarrow \quad x(t) = A \cdot \cos(\omega \cdot t + \varphi) \qquad \omega = \sqrt{\frac{D}{m}}$$

energy conservation:

$$E = \frac{p^2}{2m} + \frac{D}{2} \cdot x^2 = cst$$

turning points:

$$A = \sqrt{\frac{2 \cdot E}{D}}$$



time independent Schrödinger eqn.

$$E\,\varphi(x) = -\frac{\hbar^2}{2m}\frac{d^2\varphi}{dx^2} + \frac{D}{2}\cdot x^2\cdot\varphi(x)$$

$$\Rightarrow \quad \boxed{\frac{d^2\varphi}{dx^2} = \left(-\frac{2mE}{\hbar^2} + \frac{m\cdot D}{\hbar^2}\cdot x^2\right)\cdot\varphi(x)}$$

task: find solutions with $\varphi(\pm\infty) = 0$

Ansatz:

$$\varphi(x) \to e^{-\frac{1}{2}\alpha\cdot x^2}$$

$$\Rightarrow \quad \frac{d^2\varphi}{dx} = \frac{d}{dx}\left(-\alpha\cdot x\, e^{-\frac{1}{2}\alpha\cdot x^2}\right) =$$

$$= -\alpha\cdot e^{-\frac{1}{2}\alpha\cdot x^2} + \alpha^2\cdot x^2\, e^{-\frac{1}{2}\alpha\cdot x^2}$$

this is a solution if

- $$\alpha^2 = \frac{m\cdot D}{\hbar^2} = \frac{m^2\cdot\omega^2}{\hbar^2} \Rightarrow \alpha = \frac{m\omega}{\hbar}$$
- $$\frac{2mE}{\hbar^2} = \alpha = \frac{m\omega}{\hbar} \Rightarrow E = \frac{1}{2}\hbar\cdot\omega$$

Remark:

$$[\alpha] = \left[\frac{1}{m^2}\right]$$

$\Rightarrow$ natural length:

$$a = \frac{1}{\sqrt{\alpha}} = \sqrt{\frac{\hbar}{m\cdot\omega}}$$

oscillator length

$\Rightarrow$ ground state wave function:

$$\boxed{\varphi_0 \propto e^{-\frac{1}{2}\frac{x^2}{a^2}}} \qquad a = \sqrt{\frac{\hbar}{m \cdot \omega}}$$

$$\boxed{E_0 = \frac{1}{2}\hbar\omega} \qquad \text{zero point energy}$$

zero point energy: $E = 0$ $(x = 0)$ is impossible due to Heisenberg:

$$\Delta x \equiv b \qquad \Rightarrow \qquad \Delta p \gtrsim \frac{\hbar}{b}$$

$$\Rightarrow \quad \langle H \rangle \cong \frac{1}{2m}\frac{\hbar^2}{b^2} + \frac{m \cdot \omega^2}{2} b^2$$

minimum at $\quad b^2 = \frac{\hbar}{m\omega} \qquad \Rightarrow b = \left(\frac{\hbar}{m \cdot \omega}\right)^{1/2}$

$$\langle H \rangle \approx \hbar \cdot \omega$$

Normalization

$$\varphi_0 = N \cdot e^{-\frac{1}{2}\frac{x^2}{a^2}}$$

$$1 \equiv \int_{-\infty}^{\infty} |\varphi_0(x)|^2 \, dx = N^2 \int_{-\infty}^{\infty} dx \; e^{-\frac{x^2}{a^2}} = N^2 \cdot a \underbrace{\int d\xi \; e^{-\xi^2}}_{\sqrt{\pi}}$$

$\xi = \frac{x}{a}$

$$\Rightarrow \qquad N = \frac{1}{\pi^{1/4} \cdot \sqrt{a}}$$

$$\varphi_0(x) = \frac{1}{\pi^{1/4} \cdot \sqrt{a}} \cdot e^{-\frac{1}{2}\frac{x^2}{a^2}} \qquad \text{with} \qquad a = \sqrt{\frac{\hbar}{m \cdot \omega}}$$

Excited states

$$\varphi_n(x) \propto H_n\left(\frac{x}{a}\right) e^{-\frac{1}{2}\frac{x^2}{a^2}} \qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right)$$

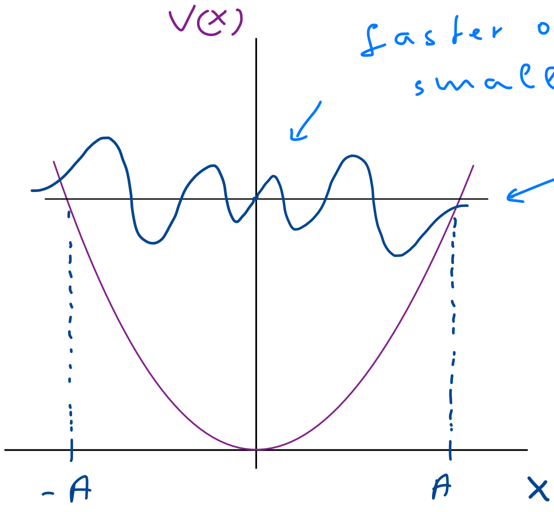
- $H_n(\xi)$ Hermite polynomials

$$H_{n+1}(\xi) = 2\xi\cdot H_n(\xi) - H_n'(\xi)$$

$$H_0 = 1$$

$$H_1 = 2\xi$$

$$H_2 = 4\xi^2 - 2$$

- 

faster oscillation $\left(p \sim \frac{h}{\lambda}\right)$
smaller amplitude

$E_7 = \hbar\omega\left(7+\frac{1}{2}\right)$

↑ seven nodes!

- Normalization factor

$$N_n = \frac{1}{\pi^{1/4}\cdot\sqrt{a}} \frac{1}{2^{n/2}\cdot\sqrt{n!}}$$

- φ_n's are orthonormal:

$$\int_{-\infty}^{\infty} dx \, \varphi_n(x)\, \varphi_m(x) = \delta_{n,m} \quad !$$

generic!

Oscillations ?

time dependent solutions

$$\psi_n(x,t) = \varphi_n(x) \cdot e^{-i\frac{E_n}{\hbar}\cdot t}$$

a linear combination of these is also a solution:

$$\boxed{\psi(x,t) = \sum_{n=0}^{\infty} c_n \, \psi_n(x,t)}$$

complex coefficients

ψ is properly normalized if $\boxed{\sum_{n=0}^{\infty} |c_n|^2 = 1}$

$$\int |\psi(x,t)|^2 dx = \sum_{n=0}^{\infty} |c_n|^2 = 1$$

Consider, e.g.:

$$\psi(x,t) = \frac{1}{\sqrt{2}} \psi_0(x,t) + \frac{1}{\sqrt{2}} \psi_1(x,t)$$

$$\psi_0(x,t) = \frac{1}{\pi^{1/4}} \frac{1}{\sqrt{a}} e^{-\frac{x^2}{2\cdot a^2}} e^{-i\frac{1}{2}\omega\cdot t}$$

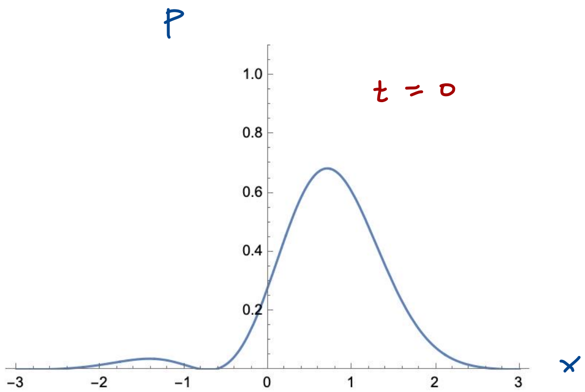
$$\psi_1(x,t) = \frac{1}{\pi^{1/4}} \frac{1}{\sqrt{a}} \frac{1}{\sqrt{2}} \frac{2x}{a} e^{-\frac{x^2}{2a^2}} \cdot e^{-i\frac{3}{2}\omega\cdot t}$$

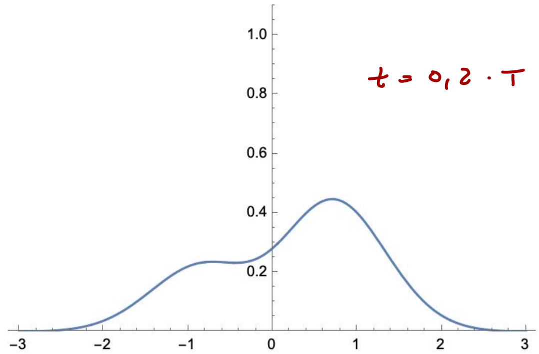
$$\Rightarrow \quad \psi(x,t) = \frac{1}{\sqrt{2}} \frac{1}{\pi^{1/4}} \frac{1}{\sqrt{a}} e^{-\frac{x^2}{2\cdot a^2}} e^{-i\frac{1}{2}\omega\cdot t} \left(1 + \frac{\sqrt{2}\cdot x}{a} e^{-i\omega\cdot t}\right)$$

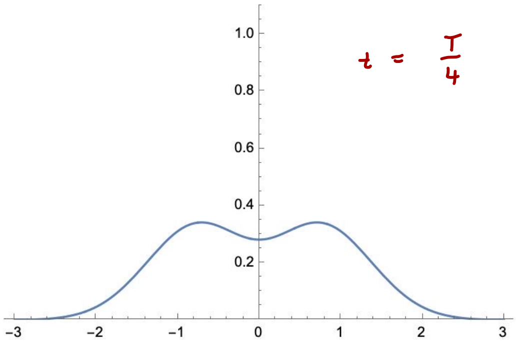
$$\rho(x,t) = |\psi(x,t)|^2$$

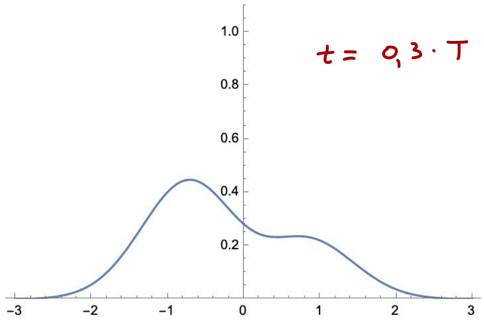
$$\rho(x,t) = \frac{1}{\sqrt{\pi}} \frac{1}{2a} e^{-\frac{x^2}{a^2}} \left(1 + \frac{2x^2}{a^2} + 2\frac{\sqrt{2}x}{a} \cdot \cos(\omega t)\right)$$

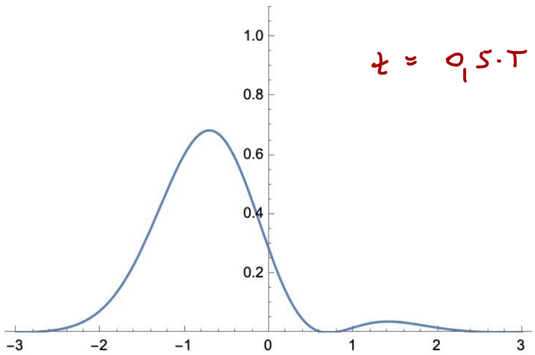
Units: $a = T = 1$ $\quad (\omega = 2\pi)$











Tunneling and scattering

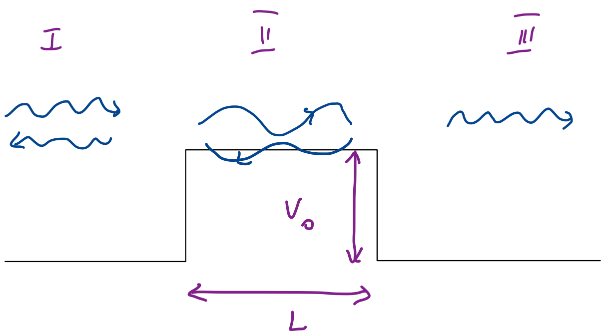
consider wave: $\psi \sim A \cdot e^{ik\cdot x} \quad e^{-i\frac{\hbar k^2}{2m}\cdot t}$

"current density"

$$J(x) = \frac{\hbar}{2mi}\Big(\underbrace{\psi^* \frac{\partial\psi}{\partial x}}_{|A|^2\cdot ik} - c.c.\Big) = |A|^2 \frac{\hbar\cdot k}{m} = v\cdot |A|^2$$

$n_\rightarrow = |A|^2 \iff$ density of incident particles

Consider barrier



Assume $E = \frac{\hbar^2 k^2}{2m} > V_0$

Region I

$$\frac{d^2\varphi}{dx^2} + \underbrace{\frac{E\cdot 2m}{\hbar^2}}_{k^2}\cdot\varphi = 0 \Rightarrow \varphi_I(x) = \underset{\text{incident}}{A\, e^{ix\cdot k}} + \underset{\text{reflected}}{B\, e^{-ix\cdot k}}$$

Region II

$$\frac{d^2\varphi}{dx^2} + \underbrace{\frac{(E-V_0)2m}{\hbar^2}}_{"q^2"}\varphi = 0 \Rightarrow \varphi_{II} = C\, e^{ix\cdot q} + D\, e^{-ix\cdot q}$$

Region III:

$$\Rightarrow \varphi_{III} = \underset{\text{transmitted}}{F\, e^{i\cdot k\cdot x}} + \cancel{G\, e^{-ik\cdot x}}$$

Boundary conditions

$$\varphi_{I}(0) = \varphi_{II}(0) \; ; \quad \varphi'_{I}(0) = \varphi'_{II}(0)$$

$$\varphi_{II}(L) = \varphi_{III}(L) \; ; \quad \varphi'_{II}(L) = \varphi'_{III}(L)$$

$\Rightarrow$ one can express B, C, D, E in terms of A

$\Rightarrow$ Reflection/transmission probability:

$$\boxed{\mathcal{R} = \frac{|B|^2}{|A|^2}} \; , \quad \boxed{\mathcal{T} = \frac{|E|^2}{|A|^2}}$$

long but simple algebra:

$$\boxed{\mathcal{T}(E>V_0) = \left(1 + \frac{v_0^2}{4\,\epsilon(\epsilon - v_0)} \sin^2\left(\sqrt{\epsilon - v_0}\right)\right)^{-1}}$$

with $v_0 = \dfrac{V_0}{\hbar^2/2mL^2}$, $\epsilon = \dfrac{E}{\hbar^2/2mL^2}$

$$\boxed{\mathcal{R} = 1 - \mathcal{T}}$$ particle conservation...

- Classically $\quad \mathcal{T} = \begin{cases} 1 & \text{for } E > V_0 \\ 0 & \text{for } E < V_0 \end{cases}$!

Case $E < V_0$

$$E - V_0 < 0 \Rightarrow \frac{d^2\varphi_{II}}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2}\varphi_{II} = 0$$

$$\Rightarrow \varphi_{II} = B e^{\varkappa \cdot x} + C e^{-\varkappa \cdot x}$$

$$\varkappa = \frac{1}{\hbar}\sqrt{2m(V_0 - E)}$$

...

$$\mathcal{T}(E<V_0) = \left(1 + \frac{v_0^2}{4\epsilon(\epsilon - v_0)}\, sh^2\left(\sqrt{v_0 - \epsilon}\right)\right)^{-1}$$

$T(\epsilon)$

$E = V_0 + \frac{\hbar^2}{2mL^2} n^2 \cdot \pi^2$

$v_0 = 50$

ϵ

Q.M. $\tau < 1$ even for $E > V_0$

$\sim$ can be reflected

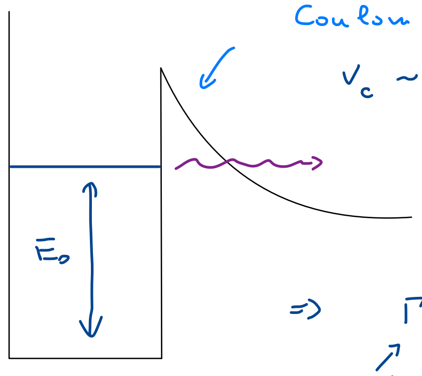
$\tau > 1$ for $E < V_0$ $\rightarrow$ tunneling!

for $V_0 >> E$:

$$\tau \approx \frac{16E}{V_0} e^{-2 \cdot L \cdot \sqrt{\frac{2mV_0}{\hbar^2}}}$$

Gamow factor

α - decay



Coulomb barrier

$$V_c \sim \frac{(Ze)\cdot 2e}{4\pi\varepsilon_0}\cdot\frac{1}{a} \sim 2Z\cdot 10^6\ eV = 2\cdot Z\ MeV$$

a: $1\ fm = 10^{-5}\ \mathring{A}$

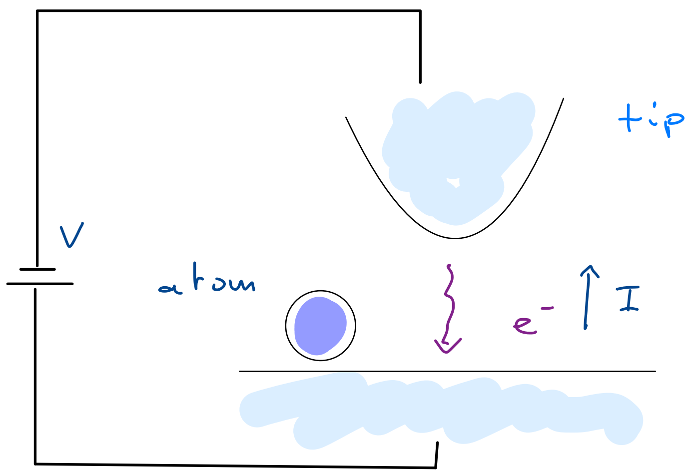
$$\Rightarrow \quad \Gamma \propto \frac{1}{\hbar}\cdot E_0\, e^{-2\cdot\int_{\text{barrier}} dx\, \frac{1}{\hbar}\sqrt{2m(V(x)-E_0)}}$$

Γ: rate of tunneling ~ decay

$\Rightarrow$ excited states decay faster

STM

Γ is exponentially sensitive to distance



- keep current fixed
  $\rightarrow$ monitor position of tip
- keep position fixed
  measure $I(V)$